TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Dismantling and Reassembling of Different Parts of Knitting Machinery

Needle cylinder service, cam-box strip and sinker-ring renewal on the circular machine: removal in sequence, part-by-part inspection against limits, reassembly with settings restored, and the fabric test that passes or fails the rebuild.

FM-4202 · ADVANCED KNITTING SESSIONAL · EXPERIMENT 2 · FABRIC MANUFACTURING SPECIALIZATION

1. Aim and theory

Aim: to dismantle the needle cylinder assembly, a cam box and the sinker ring of the laboratory machine; to inspect needles, sinkers, cams and tricks against service limits; and to reassemble with settings restored and verified by a knitted test length.

The circular knitting machine wears in characteristic places, and each wear writes a characteristic fault (the knitting counterpart of FM-4102 Experiments 2–3): a **bent or broken needle butt** gives a wale line of distorted loops and, at the limit, a dropped-stitch ladder; a **worn latch or hook** gives missed stitches and holes; a **grooved cam track** gives needle-line marks and uneven loops; a **worn trick** lets needles wobble — variable loops along single wales; **sinker wear** gives press-off and knock-over faults. Dismantling is how these are found *before* they are printed into a fabric roll. The standing rules of FM-4102 Experiment 2 apply unchanged: one zone at a time, parts in removal order, settings recorded before disturbance, nothing forced.

2. Apparatus and tools

- Circular machine isolated; cylinder and dial (as fitted) released for service; spare needle and sinker sets of the correct gauge and type; cam-box spares.
- Needle extraction hooks and pliers, socket/Allen sets, feeler gauges, magnifier or bench microscope, lint-free cloths and cleaning solvent, approved needle oil, soft brushes, parts trays and zone labels, camera, service manual with limits.

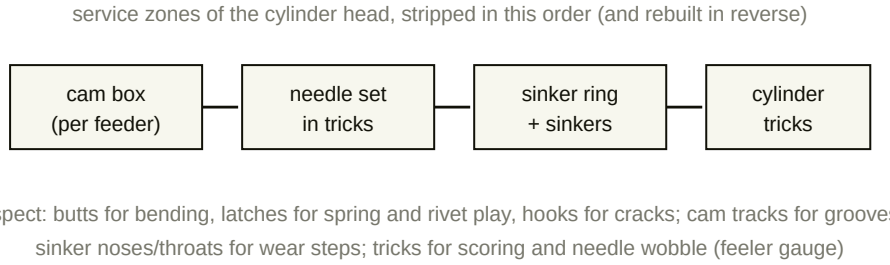


Figure 1 — The four service zones of the knitting head and their inspection points. Work proceeds zone by zone so that a half-finished machine is never more than one zone from runnable.

3. Procedure

1. **Record before stripping:** the machine's settings for the zone (stitch-cam positions at each feeder to be touched, feeder tape marks), the fault history that prompted service, and a short knitted sample of the machine's current fabric, kept for comparison.
2. **Cam box strip:** remove one feeder's cam box; lay cams, springs and fasteners in order; clean; inspect tracks under the magnifier — a track you can catch a fingernail in is a track that is marking needles; check butt-passage clearances; renew or dress parts per the manual's limits; rebuild with fasteners torqued evenly.
3. **Needle service:** work round the cylinder in marked sectors. Extract needles with the hook tool (never lever against the hook); inspect each — butt straight, latch opening freely and closing with spring, spoon undamaged, hook sound, stem straight. Segregate rejects. Refit with correct-pattern needles in the correct trick order (on patterned machines, butt-height order is a setting, not a detail); oil lightly as specified.
4. **Sinker ring:** withdraw sinkers sector by sector; inspect noses and throats for wear steps; renew worn sinkers in matched sectors; refit and check freedom of every sinker in its slot.
5. **Tricks:** with needles out of a sector, clean the tricks (fluff packing at trick bottoms is the hidden cause of "mystery" tight needles); gauge needle freedom — a needle must slide under its own weight yet not rock.
6. **Verify:** hand-turn through several revolutions watching the serviced zones; run dry briefly; then knit a test length on the standard yarn and examine it wale by wale against the pre-service sample: lines gone or accounted for, stitch length re-measured and settings trimmed to the quality sheet (Experiment 1 values).

4. Worked calculation (model data)

> **NEEDLE CENSUS OF THE SERVICED CYLINDER**

Cylinder needle count (Experiment 1): **2262 needles**.
Inspection found: bent butts 9, latches failed (no spring / rivet play) 4, hook cracked 1 — total rejects **14 needles** = 14 ÷ 2262 × 100 ≈ **0.62%** of the set.
Interpretation the viva expects: 14 defective needles are 14 defective *wales* — every one prints a line in every metre the machine knits until it is changed. The census converts a vague "the fabric had lines" into a count, a percentage and a purchase order — and re-inspection intervals are set from how fast the reject rate grows between services.

5. Inspection record (model data)

PART / ZONE	CHECK	FINDING (MODEL)
Needles (cylinder)	Butt, latch, hook, stem per needle	14 rejects of 2262 (0.62%) → replaced from correct spares
Cam box, feeder 12	Track groove (fingernail test + magnifier)	Stitch-cam track grooved → cam replaced; settings restored to record
Sinkers, one sector	Nose/throat wear steps	6 sinkers stepped → renewed as a matched set for the sector
Tricks	Fluff packing; needle slide/rock	Compacted lint in 40 tricks → cleaned; freedom restored
Test length	Wale-line inspection vs pre-service sample	Lines from serviced zones absent; stitch length 3.2 mm confirmed

6. Precautions

- Needles are sharp and springy: extraction with tools, never fingers in the trick line; rejected needles into a sharps tin, not the tray.
- Patterned machines: butt configurations go back *exactly* — photograph the sector before removal.
- No abrasive on cam tracks or tricks beyond the manual's allowance; a polished-flat cam is a ruined cam.
- The machine is not returned to production on a visual check alone — the knitted test length is the acceptance document.

7. Practice (viva preparation)

1. Match the fault to the part: (a) a single wale of tight, small loops; (b) random holes across the fabric; (c) press-offs at one feeder.
2. Why must sinkers be renewed as matched sets within a sector rather than singly at random?
3. A cylinder of 1470 needles yields 21 rejects at service. Compute the reject percentage and compare the machine's state with the worked example.
4. State why fluff at the bottom of tricks changes loop size in individual wales.
5. Why is the pre-service knitted sample kept?

Answer key

1. (a) A single wale: one needle restricted — tight trick (fluff) or bent stem limiting its movement. (b) Random holes: latch failures scattered through the set — loops not cleared or cast off cleanly. (c) One feeder's press-offs: that feeder's zone — sinker condition or cam setting local to it.
2. Knock-over and holding-down are shared acts: mixed old and new sinkers in one sector give uneven support heights, so loops form unevenly across the sector. Matched renewal keeps the sector's geometry uniform.
3. 21 ÷ 1470 × 100 ≈ **1.43%** — more than double the worked example's 0.62%: a machine in noticeably poorer state, arguing for a shorter service interval or a yarn/handling cause upstream.
4. Packed fluff shortens the trick's working depth: the needle cannot descend (or rise) fully, so it draws a different length of yarn — a shorter stitch in that wale, printed as a tight line every course.
5. It is the baseline evidence: without it, "improvement" after service is an impression. The test length is judged against the pre-service fabric defect for defect, zone for zone.

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